Block practice · Identities, equations and inequalities

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Where a line meets a fraction curve again

Rectangle and shape problems with inequalities

Showing a cubic has only one real root

OLevels.com SCHOOL

Practice workbook · Where a line meets a fraction curve again

Finding the other point where a tangent or normal line meets a curve whose equation is a fraction, using the point already known.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-15

1. Question 9 · part c · 4 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part c only.

9 A curve C has equation

$$y = \frac{qx - 2}{x - p} \qquad x \neq p$$

The curve crosses the y-axis at the point A.

The line l with equation $y = x + 2$ is the normal to C at A.

(a) (i) Show that $p = 1$

(ii) Find the value of q.

(7)

(b) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(5)

The line l meets C again at the point D.

(c) Find the x coordinate of D.

(4)

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28

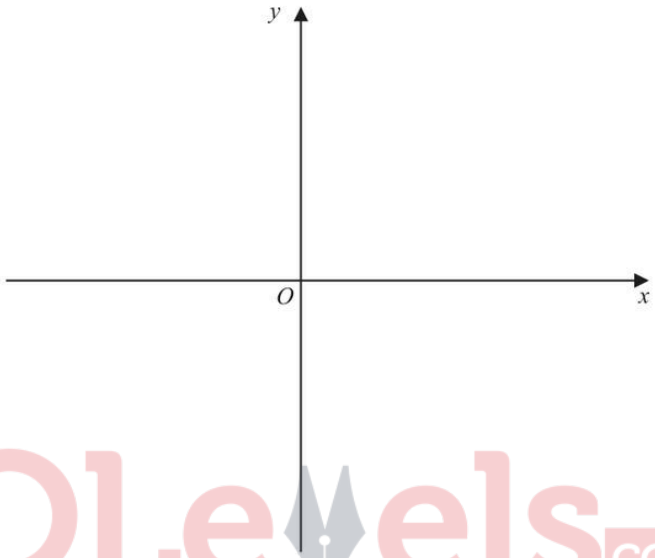
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1. Question 9 · part c · 4 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part c only.

Question 9 continued



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1. Question 9 · part c · 4 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part c only.

Question 9 continued

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1. Question 9 · part c · 4 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part c only.



1. Question 9 · part c · 4 marks · Working and answer

Working and answer

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2. Question 7 · part c · 6 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

7

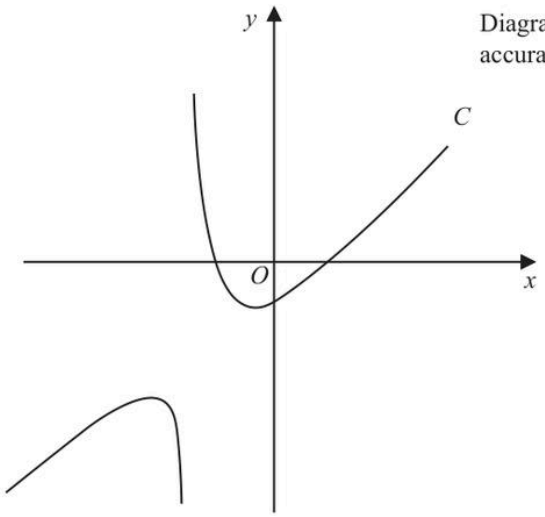


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a sketch of part of the curve C with equation

$$y = \frac{x^2 - 1}{4x + 5} \quad \text{where} \quad x \neq -\frac{5}{4}$$

(a) Write down the equation of the asymptote to C that is parallel to the y-axis.

(1)

The line l is the normal to C at the point where $x = -1$

(b) Find an equation of l

(7)

The line l meets C again at the point D

(c) Find the coordinates of D

(6)

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2. Question 7 · part c · 6 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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2. Question 7 · part c · 6 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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P 7 4 2 8 5 R A 0 1 6 3 2

2. Question 7 · part c · 6 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

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Question 7 continued

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(Total for Question 7 is 14 marks)

2. Question 7 · part c · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 11 · part d · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part d only.

11 A curve C has equation

$$y = \frac{(2a-1)x+1}{ax-6} \qquad \text{where } a \text{ is a constant and } x \neq \frac{6}{a}$$

(a) Find $\frac{dy}{dx}$

(3)

The curve crosses the y-axis at the point A.

The normal to C at the point A is the line l with equation $66y - 72x + 11 = 0$

Show that

(b) (i) $a = 3$

(4)

(ii) the equation of C is $y = \frac{5x+1}{3x-6}$ where $x \neq 2$

(1)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes with their equations and the coordinates of the points where C crosses the coordinate axes.

(5)

The line l meets C again at the point D.

(d) Find the x coordinate of D.
Give your answer as an improper fraction.

(4)

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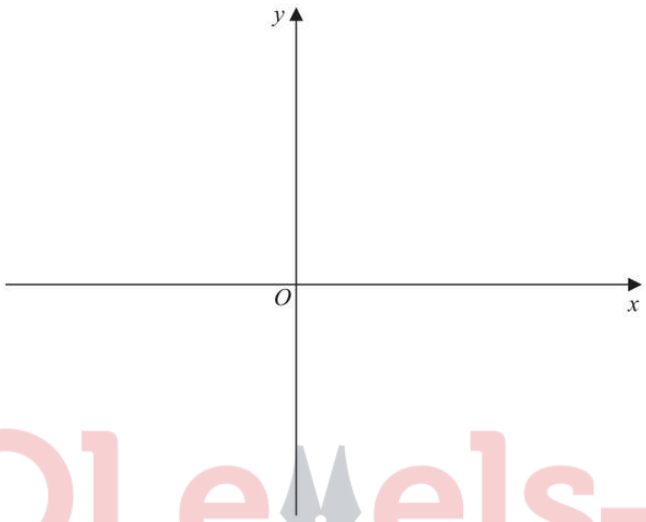
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3. Question 11 · part d · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part d only.

Question 11 continued



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3. Question 11 · part d · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part d only.

Question 11 continued

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3. Question 11 · part d · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part d only.

Question 11 continued

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3. Question 11 · part d · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part d only.

Question 11 continued

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(Total for Question 11 is 17 marks)

3. Question 11 · part d · 4 marks · Working and answer

Working and answer

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4. Question 10 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

10 The curve C with equation $y = \dfrac{6 - 3x}{x - 4}$ where $x \neq 4$, crosses the x-axis at the point P and the y-axis at the point Q

(a) Find the coordinates of

(i) P (ii) Q

(2)

(b) Write down an equation of the asymptote to C which is

(i) parallel to the y-axis (ii) parallel to the x-axis

(2)

(c) Sketch C showing clearly the asymptotes and the coordinates of the points P and Q

(3)

The line L is the normal to C at the point on C where $x = 2$

(d) Find an equation of L

(6)

The line L intersects C again at the point R

(e) Find the x coordinate of R

(3)

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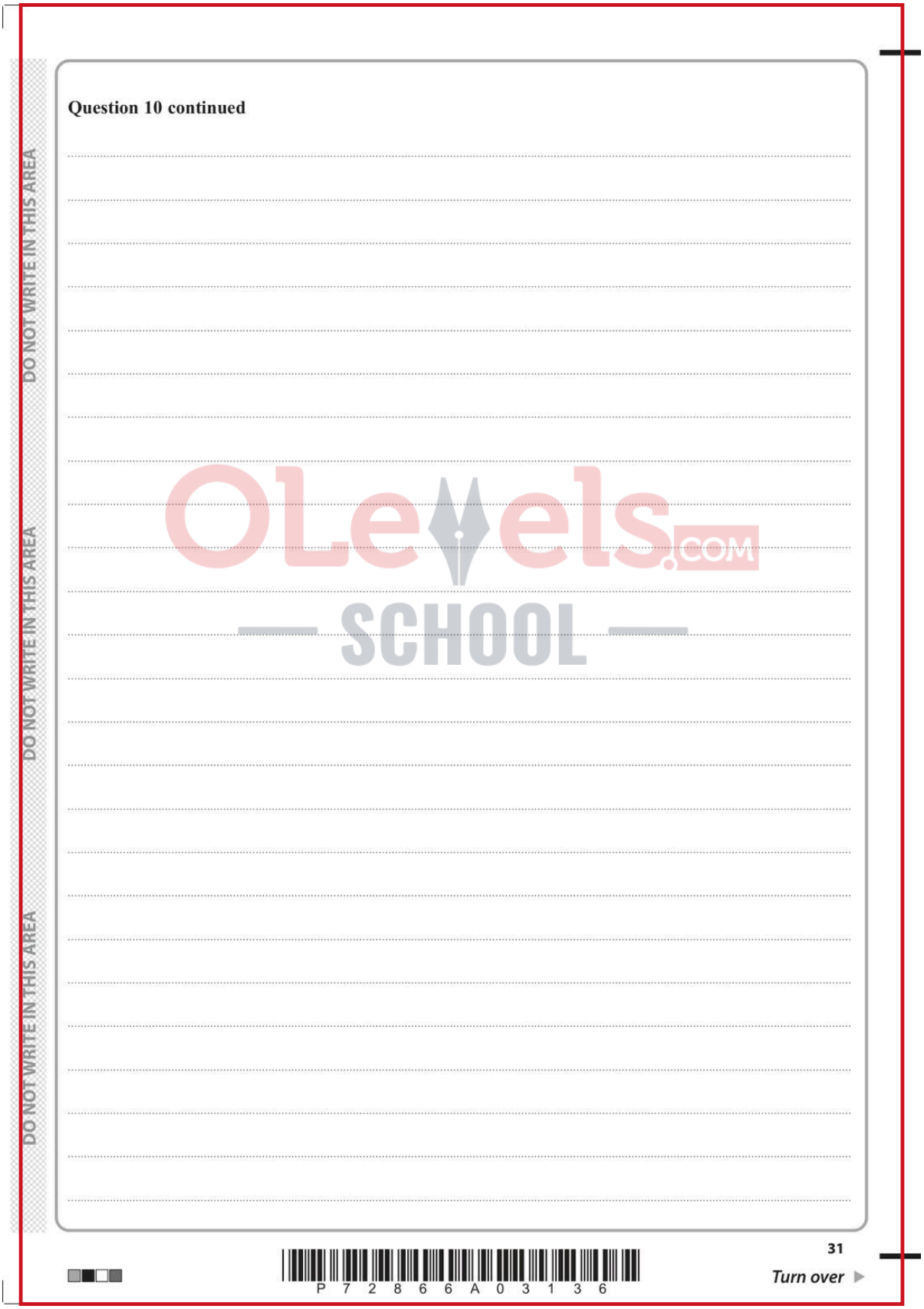
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P 7 2 8 6 6 A 0 3 0 3 6

4. Question 10 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.



4. Question 10 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

Question 10 continued

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4. Question 10 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.



4. Question 10 · part e · 3 marks · Working and answer

Working and answer

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Practice workbook · Rectangle and shape problems with inequalities

Word problems where lengths in a diagram give expressions for perimeter or area, which are then used to form and solve equations or inequalities.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-04

1. Question 2 · part whole · 6 marks

2024 NOV · Paper 02

Worked-example retry · Attempt part whole only.

2 The length of rectangle R is 2 cm greater than its width.

The area of R is greater than $8\,cm^2$ and the perimeter of R is less than 30 cm.

Given that the width of R is w cm,

find the set of possible values of w
Give your answer in the form $a < w < b$ where a and b are rational numbers.

(6)

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P 7 3 5 8 7 A 0 4 3 2

1. Question 2 · part whole · 6 marks

2024 NOV · Paper 02

Worked-example retry · Attempt part whole only.

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Question 2 continued

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(Total for Question 2 is 6 marks)

1. Question 2 · part whole · 6 marks · Working and answer

Working and answer

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2. Question 4 · part a · 2 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

4



Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a rectangle with width x metres and length $(x + 4)$ metres.

The perimeter of the rectangle is P metres and the area of the rectangle is $A\,m^2$

(a) Find, in terms of x, an expression for

(i) P (ii) A

(2)

The perimeter of the rectangle has to be less than 30 metres.
The area of the rectangle has to be greater than $12\,m^2$

(b) Find the set of possible values for x
Give your answer in the form $a < x < b$

(5)

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2. Question 4 · part a · 2 marks · Working and answer

Working and answer

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3. Question 4 · part b · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

4



Diagram **NOT** accurately drawn

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Give your answer in the form $a < x < b$

(5)

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3. Question 4 · part b · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 7 marks)

3. Question 4 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 5 · part whole · 7 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

5



Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a rectangle with width x cm and length $(2x+3)$ cm.

The perimeter of the rectangle is P cm and the area of the rectangle is A cm^2

$$P > 10 \quad \text{and} \quad A < 35$$

Find the set of possible values for x
Give your answer in the form $a < x < b$ where a and b are rational numbers.
Show clear algebraic working.

(7)

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4. Question 5 · part whole · 7 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 5 continued

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(Total for Question 5 is 7 marks)

4. Question 5 · part whole · 7 marks · Working and answer

Working and answer

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5. Question 2 · part a · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part a only.

2

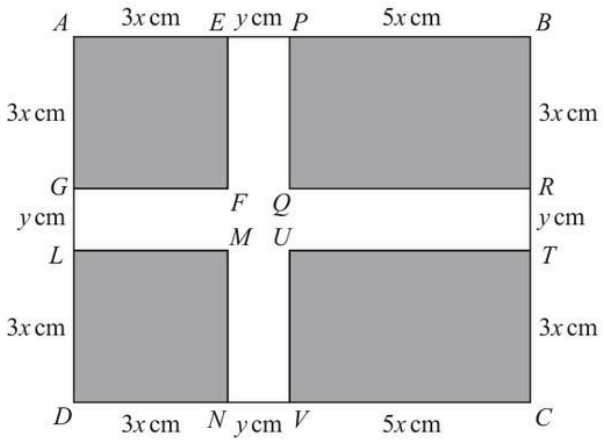


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the design for a flag consisting of a white cross on a grey background.

AEFG and *DLMN* are squares with sides of length $3x$ cm.
BPQR and *CTUV* are rectangles with sides of length $5x$ cm and $3x$ cm.
The width of the cross is y cm.

The total area of the flag is H cm^2

(a) Write down an expression, in terms of x and y, for H.

(1)

Given that the area of the cross is K cm^2

(b) show that $K = 14xy + y^2$

(3)

The total area of the flag is to be 3432 cm^2 and the area of the cross is to be 1080 cm^2

(c) Find the value of x and the value of y

(5)

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P 7 1 6 6 1 A 0 4 3 2

5. Question 2 · part a · 1 mark · Working and answer

Working and answer

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6. Question 2 · part b · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.

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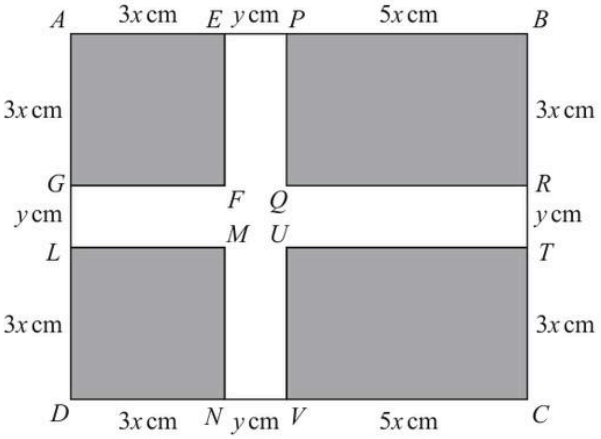


Diagram **NOT** accurately drawn

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(5)

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6. Question 2 · part b · 3 marks · Working and answer

Working and answer

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Practice workbook · Showing a cubic has only one real root

Proving that a cubic equation has exactly one real root by showing its quadratic factor has a negative discriminant.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-12

1. Question 4 · part b · 3 marks

2022 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part b only.

4 $f(x) = x^3 + px^2 + qx + 7$ where p and q are integers.

$(x + 1)$ is a factor of $f(x)$
The remainder when $f(x)$ is divided by $(x + 2)$ is -5

(a) Find the value of p and the value of q

(5)

(b) Hence, show that $f(x) = 0$ has only one real root.

(3)

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1. Question 4 · part b · 3 marks

2022 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 8 marks)

1. Question 4 · part b · 3 marks · Working and answer

Working and answer

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2. Question 5 · part b · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

5 $f(x) = x^3 + x^2 + x + c$ where c is a constant

The remainder when $f(x)$ is divided by $(x-2)$ is 4 times the remainder when $f(x)$ is divided by $(x+1)$

(a) Show that $c = 6$

(4)

Given that $(x+2)$ is a factor of $f(x)$

(b) show that the equation $f(x) = 0$ has only one real root.

(4)

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2. Question 5 · part b · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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Question 5 continued

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(Total for Question 5 is 8 marks)

2. Question 5 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 1 · part b · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

1 $f(x) = 6x^3 - 13x^2 + ax - 10$ where a is a constant

Given that $(3x - 2)$ is a factor of $f(x)$

(a) show that $a = 21$

(2)

(b) Hence show algebraically that the curve $y = f(x)$ has only one intersection with the x-axis.

(4)

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(Total for Question 1 is 6 marks)

3. Question 1 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 10 · part d · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

10 (a) Use the factor theorem to show that $(4x - 3)$ is a factor of

$$16x^3 + 11x - 15$$

(2)

(b) Using formulae given on page 2, show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

(c) Show that the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

becomes the equation

$$16x^3 + 11x - 15 = 0$$

by using the substitution $x = \cos\theta$

(4)

(d) Hence show that any solution of the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

is given by $\cos\theta = \frac{3}{4}$

(4)

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4. Question 10 · part d · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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4. Question 10 · part d · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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4. Question 10 · part d · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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(Total for Question 10 is 15 marks)

4. Question 10 · part d · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 1 · part b · 5 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

1 $f(x) = (x - 3)[x^2 + (p - 2)x + q]$

Given that $f(0) = -12$

(a) find the value of q.

(2)

(b) Find the range of values of p for which the cubic equation $f(x) = 0$ has only one real root.

(5)

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(Total for Question 1 is 7 marks)

5. Question 1 · part b · 5 marks · Working and answer

Working and answer

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